

Module 6:

IMINT



Topic 1: IMINT



Introduction to IMINT Techniques

Reverse Image Searching:

There are a lot of tools used to perform reverse image searching, such as:

<https://images.google.com>

<https://www.bing.com/images>

<https://tineye.com>

<https://lens.google>

<https://yandex.com/images>

Introduction to IMINT Techniques

Metadata:

In general, metadata is the data that describes/ provides some information about other data.

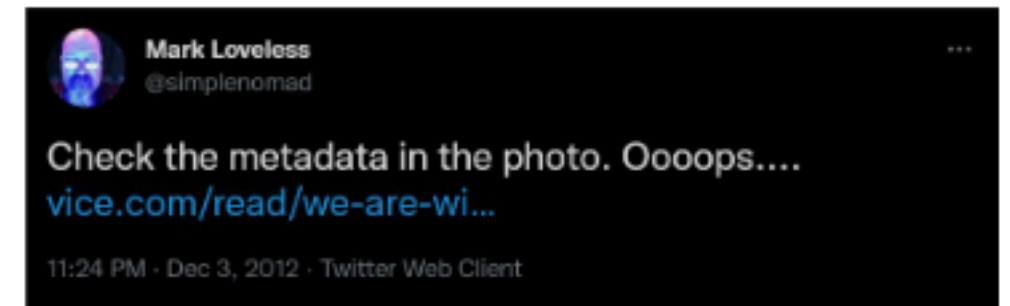
Some metadata examples:

- Time and date of creation, modification and access.
- Creator/ Author of the file.
- Location where the file was created.
- Programs used to create/ modify the file.

Introduction to IMINT Techniques

In November 2012, John McAfee (the founder of McAfee antivirus) had been a person of interest in a murder case in Belize (a Country in Central America).

In December 2012, the location of John was revealed when Vice (a Canadian-American magazine) posted a photo with the geotags metadata still attached.



Introduction to IMINT Techniques

I Know Where Your Cat Lives (<https://iknowwhereyourcatlives.com/>) is a data visualization project that locates public images tagged with cats by the geographic coordinates embedded in their metadata. Those images were collected from publicly available APIs provided by Flickr, Twitpic, Instagram, and others.



Introduction to IMINT Techniques

There are a lot of tools to check the metadata of files, such as **ExifTool**.

exiftool is used to read/write metadata in files.

- CLI: <https://exiftool.org/install.html>
- GUI: <https://exiftool.org/gui/>
- Web: <https://exif.tools/> | <https://exifmeta.com/>

IMINT and GEOINT

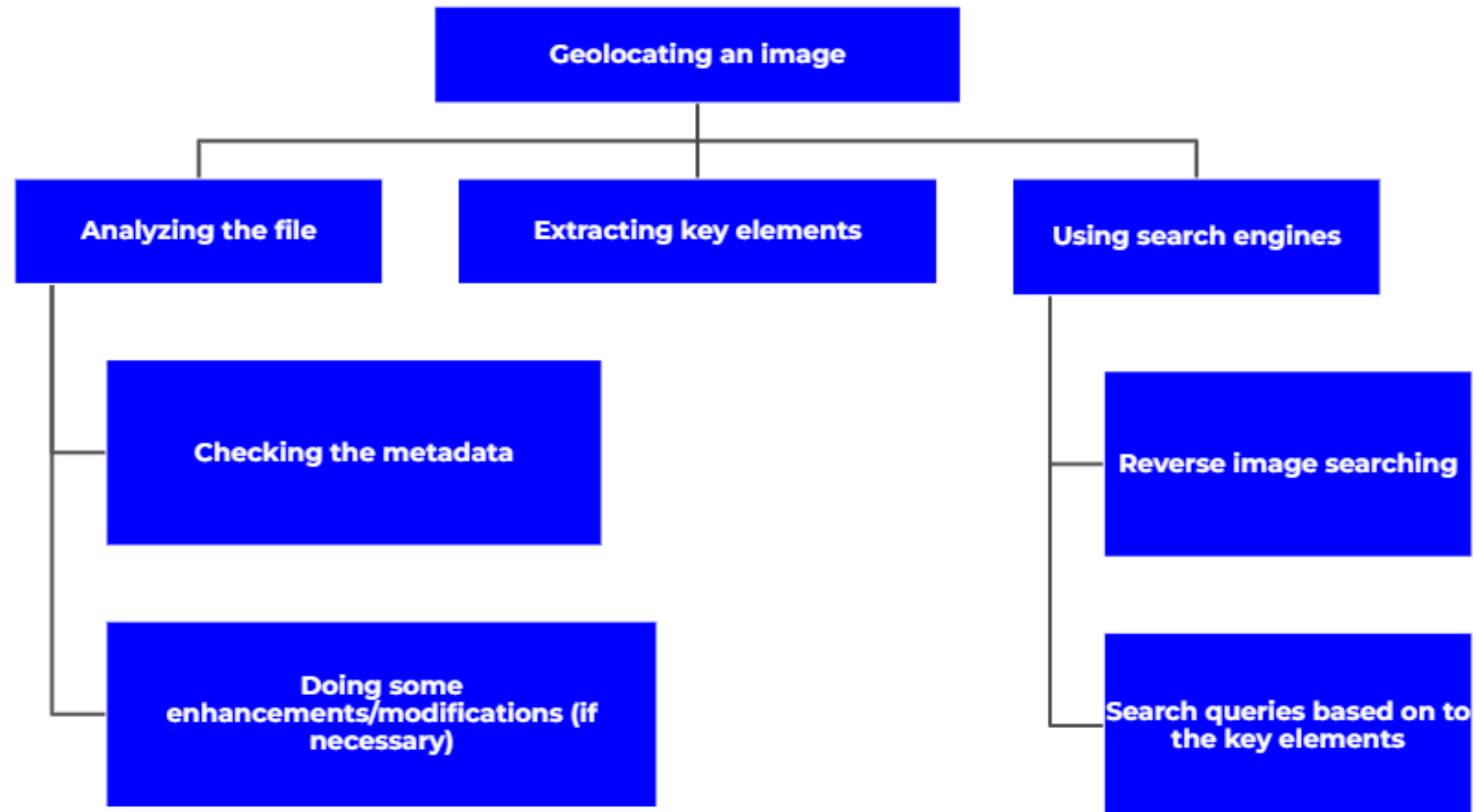
“GEOINT data” is not a new or special type of data. The term simply refers to any data used to create GEOINT, which can be derived from multiple classified or unclassified sources. Since geospatial data is the primary source for all three major elements of GEOINT—geospatial information, imagery, and imagery intelligence—it is considered the main source of data for GEOINT.

Just as other intelligence disciplines use GEOINT to develop a comprehensive response to an intelligence problem, the GEOINT discipline incorporates data from other intelligence disciplines, such as Human Intelligence (HUMINT), Signals Intelligence (SIGINT), Measurement and Signatures Intelligence (MASINT), and Open-Source Intelligence (OSINT).

- [Source: Geospatial Intelligence \(GEOINT\) Basic Doctrine Publication 1-0 \(September 2006\) - National Geospatial-Intelligence Agency](#)

IMINT and GEOINT

There are several things that can be done if you want to geolocate an image, **for example**:



IMINT and GEOINT

Checking the file's metadata:

The following online tools can be used to check the metadata:

- <https://exif.tools/>
- <https://www.pic2map.com/>
- <https://www.verexif.com/en/>

Note: It's not always guaranteed to find the geotags metadata attached to images.

IMINT and GEOINT

Checking the file's metadata | Example:



IMINT and GEOINT

Doing some enhancements/modifications on the image (if necessary):

Sometimes you may need to do some enhancements/modifications on the image to be able to analyze it properly, such as: flipping, rotating, cropping, modifying the brightness level, ... etc.

There are many tools to do that, such as:

<https://www.photopea.com/>



IMINT and GEOINT

Extracting the key elements from the image:

A key element (in the case of geolocating images) is something in the image that helps in identifying the location or narrowing the search into a more specific area.

For example: Unique landmarks, architectural styles, storefront signs, road signs, license plates, and linguistic clues.

IMINT and GEOINT



Source: <http://www.worldlicenseplates.com/>

IMINT and GEOINT



Australia



Qatar



Japan



Singapore

IMINT and GEOINT



Source: <https://imgur.com/gallery/K9sBw4T>

IMINT and GEOINT

Extracting the key elements from the image: | Example:



Lab

Series



Video Analysis

Video Analysis:

Videos consist of frames, and a frame is one of the many still images which compose the complete moving picture.

So, videos' frames can be analyzed individually, and as it is difficult to analyze huge number of frames, it is better to focus on the keyframes that contain key elements related to the case, for example, a frame that reveal some key elements will be more helpful in geolocating a video rather than a blurry frame that does not reveal anything.

Video Analysis

<http://www.watchframebyframe.com/> is an online tool to watch YouTube videos frame by frame and in slow motion

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Watch YouTube and Vimeo videos frame by frame and in slow motion

Video Analysis

FFmpeg is a collection of libraries and tools to process multimedia content such as audio, video, subtitles and related metadata.

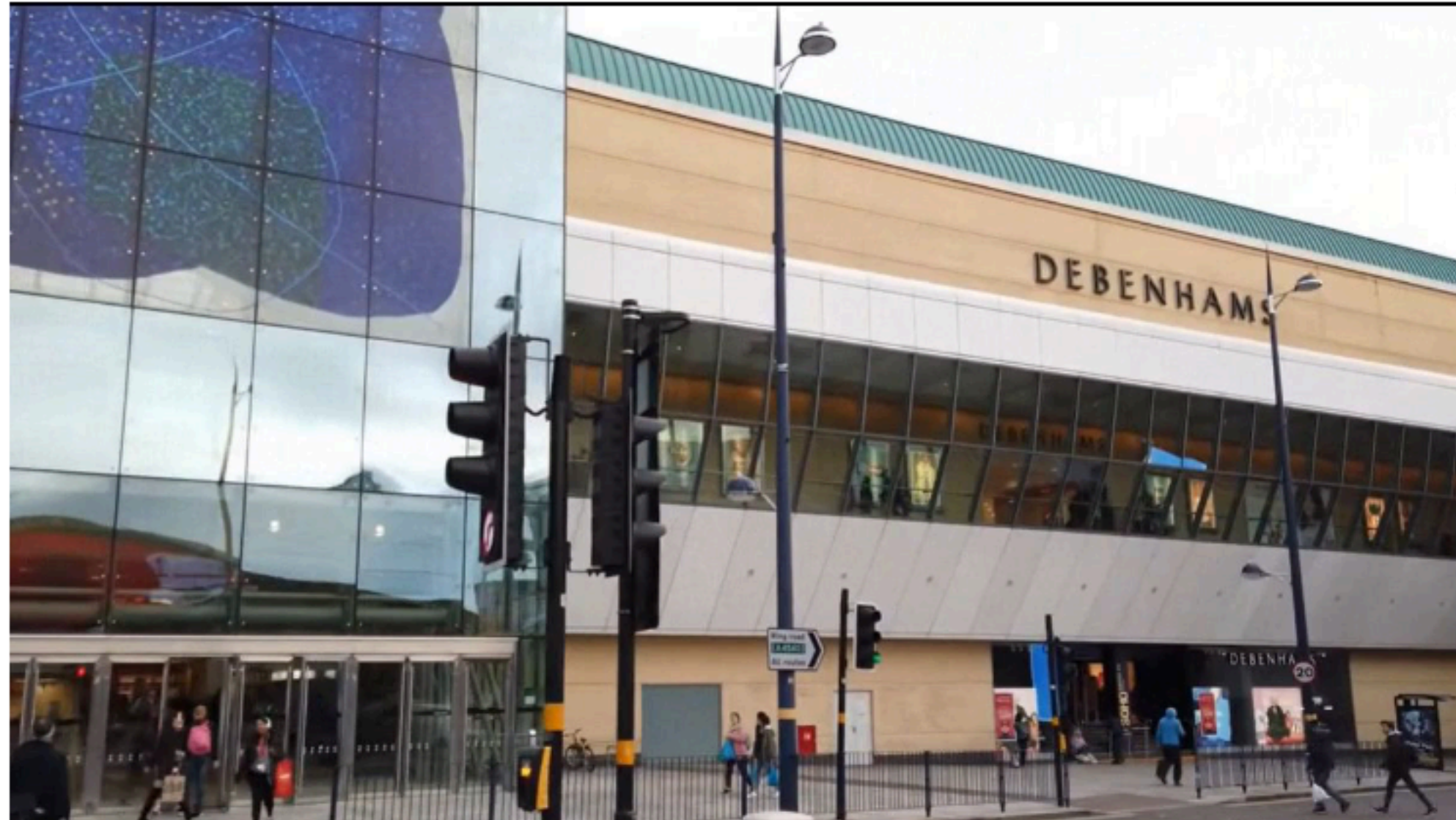
We can use it to extract the frames of a video, for example:

```
ffmpeg -i Video.mp4 Frame%08d.png
```



Video Analysis

Video Analysis | Example:



Source: <https://virtualvacation.us/guess>